

LESSON 3-6 • SESSION 2 • PRACTICE • FORM B**Metric Unit Conversions**

Learning target – I can convert between metric units by multiplying or dividing by powers of ten. · 6.RP.A.3d

REMEMBER

- Metric units use **powers of 10**: $1 \text{ km} = 1,000 \text{ m}$ · $1 \text{ m} = 100 \text{ cm}$ · $1 \text{ cm} = 10 \text{ mm}$ · $1 \text{ kg} = 1,000 \text{ g}$ · $1 \text{ L} = 1,000 \text{ mL}$.
- Big unit to small unit: **multiply** ($3 \text{ kg} = 3,000 \text{ g}$).
Small to big: **divide** ($4,000 \text{ g} = 4 \text{ kg}$).
- Make the units **match** before you divide or compare.

WORKED EXAMPLE

A bag has 2.4 kg of flour. How many 300 g portions can you make?

Step 1 Match units: $2.4 \times 1,000 = 2,400 \text{ g}$.**Step 2** Divide: $2,400 \div 300 = 8$.**Step 3** Check: $8 \times 300 = 2,400 \text{ g}$.**Answer 8 portions****WORDS** **metric units** (*unidades métricas*) · **powers of 10** (*potencias de 10*) · **convert** (*convertir*)**PRACTICE***show your work · label units***1.** Convert each measure.

a. $9 \text{ km} = \underline{\hspace{1cm}} \text{ m}$

b. $2,500 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$

c. $6 \text{ kg} = \underline{\hspace{1cm}} \text{ g}$

d. $175 \text{ cm} = \underline{\hspace{1cm}} \text{ m}$

e. $3,820 \text{ m} = \underline{\hspace{1cm}} \text{ km}$

f. $4.5 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$

2. Juice Dispenser. A dispenser has 6 L of juice. Complete the table. How many 250 mL cups can it fill?**Hint** Divide the total milliliters by 250.

Liters	1	2	4	6
Milliliters	1,000	2,000	4,000	

Answer _____ cups**3. 6K Run.** A charity race is 6 km long. ($1 \text{ km} = 1,000 \text{ m}$)

a. Meters: _____ m

b. A runner has gone 2,700 m. Meters left: _____ m

*Show your work.***4. Lab Mass.** A student measures 425 g of salt. ($1 \text{ g} = 1,000 \text{ mg}$, $1 \text{ kg} = 1,000 \text{ g}$)

a. Milligrams: _____ mg

b. Kilograms: _____ kg

*Show your work.***5. Lab Ribbon.** A teacher cuts a 2.4-meter ribbon into 40 cm pieces. ($1 \text{ m} = 100 \text{ cm}$)**Hint** Change meters to centimeters before you divide.

a. Ribbon length in centimeters: _____ cm

b. Number of 40 cm pieces: _____ pieces

Show your work.

6. **Backpack Limit.** A backpack weighs 5,300 grams. The safety limit is 5 kg. Is the backpack under the limit? (1,000 g = 1 kg)

Hint Change grams to kilograms: divide by 1,000.

Show your work.

Yes or no _____

7. **Recycling Bin.** A full recycling bin has a mass of 9.6 kilograms. What is its mass in grams? (1 kg = 1,000 g)

Hint Kilograms to grams is big to small: multiply.

Show your work.

Answer _____ g

8. **Class Relay.** Each of 6 runners covers 750 meters in a relay.

- a. Total distance in meters: _____ m
- b. Total distance in kilometers: _____ km
- c. Which unit would you use to report the total? Why? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Maya's Work.** Maya converted 5 kilograms to grams by dividing: $5 \div 1,000 = 0.005$ g. Explain her mistake and find the correct mass.

TEST PRACTICE

choose the best answer

10. 250 centimeters equals how many meters?
- A 25,000 B 2.5 C 25 D 0.25
11. 3 kilograms equals how many grams?
- A 30 B 300 C 3,000 D 0.003

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

24	2.5	4.5	0.425	3.82	42.5	3,300	960	382	9,600
T	M	S	I	E	X	R	Q	Z	C

#1b	#1e	#2	#3b	#4b	#7	#8b

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____